

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 21

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• B (about 40:40:20 sand:silt:clay) • Balanced mix gives good drainage and aeration from the sand fraction • Water retention and nutrient (ion) retention from the clay fraction; easy root penetration and cultivation	3
01.2	• Large sand particles leave large pore spaces so water drains rapidly (high permeability / aeration) • Nutrient ions adsorb onto the (charged) surfaces of clay particles but not onto sand • With only 5% clay there is little surface to hold ions so nutrients are leached; low water retention too	3
01.3	• C (clay) holds much water so has a high thermal capacity: warms and cools slowly; A warms rapidly after cold weather • Clay particles held together by capillary water so roots find it hard to push between them; penetration is easy in sandy A	2
01.4	• Shake a known mass of soil with water in a measuring cylinder and leave to settle; particles settle in order of size (sand first, then silt, then clay; organic matter floats) • Measure the depth / volume of each layer to calculate the percentage of each fraction, then use the soil triangle to name the textural class	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-160, 412; AQA spec 3.2.5.1.	
02.1	• Spearman's rank correlation coefficient • Chi-squared test (frequencies / counts in categories) • t-test (two means of measured data) • Spearman's rank correlation coefficient	4
02.2	• Chi-squared compares frequencies / counts in categories; it cannot be used for continuously variable measured data	1

Question	Marking guidance	Marks
02.3	- The value (ignoring sign) exceeds the critical value so the correlation is significant at the 5% level - The negative sign shows a negative correlation: one variable decreases as the other increases - There is less than a 5% probability the correlation arose by chance; reject the null hypothesis (but correlation is not proof of causation)	3
02.4	- Show the standard deviation / range of the original values around each mean - If the bars of two means overlap greatly the difference is probably not significant; little overlap suggests a significant difference	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	
03.1	- Lowest overnight (about 25 GW at 04:00), rising steeply from 06:00 to about 40 GW by 08:00 - Fairly steady through the day then a peak of about 50 GW at 18:00 (evening) before falling overnight	2
03.2	24-hour day / night cycle: people asleep overnight; morning rise as heating, lighting, cooking, industry and offices start - Evening peak: lighting, cooking, heating as people return home; short-term 'TV pickup' at programme breaks / mealtimes - Other causes: seasonal (winter heating / lighting, summer air conditioning); weather-related (cold snaps); weekday / weekend differences in industrial use	4
03.3	- They generate 24 hours a day because it is uneconomic / slow to turn them off and on, and output changes slowly - Demand falls faster at night than output can be reduced, so the surplus must be stored or is lost as heat from cables	2
03.4	<i>Any 2 from:</i> - Intermittent renewable resources: wind, solar, tidal - Bulk delivery of transported fuels (oil tankers, coal trains, biofuel harvests) - Breakdowns / maintenance of power stations	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 221-222; AQA spec 3.3.4.1.	
04.1	- Trees absorb CO₂ by photosynthesis and store carbon in wood as they grow - Once mature there is no further net storage (photosynthesis balanced by respiration / decomposition), though the standing forest remains a reservoir - Harvesting the timber and keeping it as wood (eg house construction) then replanting produces additional storage as the new trees grow	3

Question	Marking guidance	Marks
04.2	• Pre-combustion: the fuel is modified before use so no CO₂ is released when burnt - eg gasification of coal to hydrogen and CO₂ (CO₂ stored, hydrogen distributed); oxy-fuel combustion in pure oxygen giving only CO₂ and water vapour which is condensed out • Advantage: capture at a large central facility; suits small dispersed sources (vehicles) that cannot capture CO₂ • Post-combustion: CO₂ removed from exhaust gases by solvents, membrane filtration, adsorption / desorption, cryogenic separation, graphene • Disadvantage: only about 20% of flue gas is CO₂ so separation is costly / energy-intensive; none fully developed	4
04.3	<i>Any 3 from:</i> • Injected into depleted oil / gas fields or saline aquifers • Injected into oil reservoirs to provide pressure for secondary oil recovery • Reaction with fractured basalt to form solid carbonate minerals • Concern: leakage back to the atmosphere / monitoring over centuries; limited suitable geology; cost	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 151-152; AQA spec 3.2.4.2.	
05.1	• Double hull: a gap of up to 2 m between the hulls so damage to the outer hull in a collision / grounding does not release oil • Inert gas system: cooled engine exhaust gas (no oxygen) fills the space above the oil so an explosive oil vapour / air mixture cannot form • Separate ballast tanks: ballast water is not carried in oil tanks so no residual oil is discharged when ballast is pumped out • Twin engines / rudders: if one fails the ship can still be steered / propelled and is not carried onto rocks by currents or wind • Crude oil washing: sludges are mixed and removed with the cargo so tanks do not need washing with seawater that was then discharged; oily water is unloaded at the terminal	5
05.2	• Offshore routes keep tankers further from the coast so if there is engine failure there is more time to deal with it before the ship is carried ashore • GPS gives accurate navigation; AIS lets ship positions be monitored and coordinated to avoid collisions and groundings	2
05.3	• The tank stands on an impermeable base surrounded by a wall • The volume enclosed by the wall is greater than the tank volume so all the oil is contained if the tank splits / cannot overflow	2
05.4	• Separates floating oil from drainage water so the oil is retained (for later treatment) and only water flows to the drains / river	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 259-260; AQA spec 3.4.3.2.7.	
06.1	• 38.0 / 0.002 • = 19 000 (19 000 times / 1.9×10^4)	2
06.2	• Biomagnification (accept bioaccumulation as part of the process only if biomagnification is also implied by increase up the food chain)	1

Question	Marking guidance	Marks
06.3	• Persistence: the pesticide is not broken down (by light, microbes, enzymes) so it remains in the environment / organisms long enough to pass along the food chain • Liposolubility: the pesticide dissolves in lipids / fat so it is stored in body tissue and not excreted in (water-based) urine • Each organism eats many organisms from the level below and retains the pesticide so concentration rises at each trophic level • Herons have long lives so accumulate over time	4
06.4	<i>Any 3 from:</i> • Organochlorines killed non-target insects (bees, butterflies) and top predators (birds of prey, otters) through biomagnification • They persist for decades so the damage continued long after use • Use in houses (on walls / nets) does not enter food chains / no runoff into water, so does not biomagnify • Benefit of preventing malaria deaths outweighs the risk (risk:benefit)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-264, 266; AQA spec 3.4.3.2.8.	
07.1	• Igneous intrusions (batholiths) of magma cool within the crust, heating groundwater to produce superheated, pressurised water • The hot water dissolves minerals from the surrounding rock and the mineral-rich solutions travel along fissures / cracks away from the batholith • As the solutions cool the dissolved minerals crystallise in order of solubility (least soluble first) - fractional crystallisation • So different metals are separated and concentrated in veins, eg tin, copper, lead, silver, gold, arsenic (SW England batholith)	4
07.2	• Dissolved iron compounds in ancient seas were oxidised by the oxygen released by early photosynthetic organisms (2.5-1.8 billion years ago) • Iron oxides are insoluble so they were deposited as sediments (haematite, magnetite)	2
07.3	• Flowing water carries solid particles; the ability to carry them depends on the water velocity and the density of the solids • As velocity falls, dense minerals (gold, tin ore) are deposited first while lighter material is carried further, concentrating the heavy minerals in particular places	2
07.4	<i>Any 2 from:</i> • Limestone / chalk from the shells of marine organisms • Coal from terrestrial vegetation • Crude oil and natural gas from marine organisms (plankton)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 132-134; AQA spec 3.2.3.2.	
08.1	• B - Tropospheric ozone (formed by reactions of NO_x in sunlight)	1
08.2	• Primary pollutant: released directly by human activities, eg sulfur dioxide from burning coal / smelting sulfide ores; NO_x from high-temperature combustion • Secondary pollutant: produced by chemical reactions between primary pollutants, often with non-pollutants (water, oxygen, sunlight) • Sulfur dioxide is oxidised by ozone to sulfur trioxide which dissolves to form sulfuric acid; NO_x dissolve to form nitric acid • The acids (sulfurous, sulfuric, nitric) are secondary pollutants; ozone itself is a secondary pollutant of NO_x photochemistry	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 237-238; AQA spec 3.4.1.	
09.1	- Conventional: $40 / 100 = 40\%$ - CHP: $(35 + 45) / 100 = 80\%$	2
09.2	- Extracting less energy as electricity leaves the cooling / condenser water at a higher temperature - Hotter water is more useful for district heating of homes, fish farms, greenhouses	2
09.3	Hot water loses heat and needs pumping energy over distance; heat cannot be transported as far as electricity	1
09.4	- Long narrow pipes give a large surface area and copper is a good thermal conductor so heat transfers quickly - Hot and cold fluids flowing in opposite directions maintain a temperature gradient along the whole length so more heat is transferred - Slower flow allows more time for heat to transfer	3
09.5	- One large tank has a lower surface area : volume ratio than many small ones - A sphere has the smallest surface area for any volume; less surface means less heat loss (with insulation of low thermal conductivity)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 233-235; AQA spec 3.3.4.2.3.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Control of greenhouse gases: energy conservation (transport, buildings, industry), renewable and nuclear energy replacing fossil fuels, CCS, reduced deforestation and afforestation (carbon sequestration), methane capture from landfill / digesters, HFC controls, changes in agriculture • Geoengineering: white roofs, ocean fertilisation, solar shades - largely untried, possible unpredictable consequences • International agreements: Kyoto Protocol, Paris Agreement - targets, but non-binding / uneven participation; carbon trading and taxes; comparison of per capita carbon footprints between countries • Evidence of limited success: global CO₂ concentrations still rising (about 280 ppm pre-industrial to over 410 ppm); temperatures up about 1 °C; emissions in some developed countries falling but rising in developing economies • Reasons for difficulty: fossil fuels underpin economies, need for major lifestyle change, cost, time delays and natural fluctuations masking effects, political disagreement, developing countries' right to develop • Adaptation strategies (flood defences, drainage, managed retreat, infrastructure changes) reduce impacts but do not address the cause • Comparison with the Montreal Protocol's success for ozone • Judgement: partial success - technology exists and some regional reductions achieved, but global strategies have not yet stopped the rise; tipping points make urgency greater	9
10.2	• The total greenhouse gas / CO₂ emissions caused by an individual, organisation, product or country (often expressed per capita per year)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-107; AQA spec 3.2.1.2.	

Question	Marking guidance	Marks
11.1	Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: energy density, renewability, abundance, location, intermittency / predictability, storage, transport, conversions, technology level, environmental impacts • Fossil fuels: choosing lower-impact fuels (gas over coal), desulfurisation, CCS, efficient combustion, siting power stations near cooling water and coal ports; new extraction technologies (directional drilling, secondary / tertiary recovery) and their impacts (fracking, tar sands) • Nuclear: high energy density reduces transport; waste properties (half-life) determine storage; thorium and fusion prospects • Renewables: matching resource to location (wind, HEP, solar, wave, tidal, geothermal, biofuel siting); intermittency managed by storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management and diversity; low-impact designs (helical turbines, VAWT, tidal lagoons) • Energy conservation reduces demand: transport, buildings, industry (CHP, heat exchangers) • Impacts before and during use; life-cycle / embodied energy • Evaluation: no resource is impact-free; knowledge allows the best mix and mitigation; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	25

Question	Marking guidance	Marks
11.2	Discuss the strategies that can be used to provide sustainable supplies of water for a growing population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Context: uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • New sources: rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • Management: artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • Conservation: low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering; pollution control to protect sources • Treatment: matching quality to use; screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	25